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# VIDYAVARDHAKA COLLEGE OF ENGINEERING

Autonomous Institute, Affiliated to Visvesvaraya Technological University, Belagavi
Gokulam 3rd Stage, Mysuru 570 002

## Seventh Semester B.E. Examinations: December 2024 – January 2025

## Blockchain Technology

**Duration: 3 Hour** **Max. Mark: 100**

**INSTRUCTION TO STUDENTS**

1) Answer **One Full** question from each module.
2) Three module questions are compulsory and remaining two modules will have internal choice.

| Q.No | Module-I | Marks | BL | CO/PO |
|---|---|---|---|---|
| 1(a) | Define Blockchain. Explain the key features of blockchain. | 8 | L2 | CO1/PO1 |
| 1(b) | Differentiate between Proof of Work and Proof of Stake. | 6 | L2 | CO1/PO1 |
| 1(c) | Explain the difference applications of Blockchain technology. | 6 | L2 | CO1/PO1 |
| | **OR** | | | |
| 2(a) | What are the requirements which must be met to provide the desired results in a consensus mechanism? Discuss the same. | 6 | L2 | CO1/PO1 |
| 2(b) | Illustrate the Generic Elements of blockchain. | 8 | L2 | CO1/PO1 |
| 2(c) | "CAP theorem has been violated in Blockchain". Justify the statement. | 6 | L2 | CO1/PO1 |
| | **Module-II** | | | |
| 3(a) | Describe the following with respect to blockchain architecture.<br>(i) Centralized System (ii) Decentralized system | 4 | L2 | CO1/PO1 |
| 3(b) | Illustrate the various modes of operations for the following Block ciphers.<br>(i) Electronic Code book (ii) Cipher Block chaining | 8 | L3 | CO2/PO1 |
| 3(c) | Illustrate Generic cryptography models with a neat diagram. | 8 | L3 | CO2/PO1 |
| | **Module-III** | | | |
| 4(a) | Illustrate the working principle of SHA algorithm. | 10 | L3 | CO2/PO1 |
| 4(b) | Analyze RSA algorithm for a given two prime numbers 7 and 13. Generate public key and private key and encrypt the message "WELCOME". | 10 | L4 | CO3/PO2 |
| | **Module-IV** | | | |
| 5(a) | Define smart contract. Enumerate the properties of smart contract. | 6 | L2 | CO1/PO1 |
| 5(b) | Analyze how the given instruction is executed. Using P2PKH script execution.<br><sig><pubk> op_Dup op_HASH160<PubHash><br>OP_EQUAL VERIFY OP_CHECKSIG | 7 | L4 | CO3/PO2 |
| 5(c) | Analyze the properties of Ricardian contract. | 7 | L4 | CO3/PO2 |
| | **Module-V** | | | |
| 6(a) | Illustrate the following<br>(i) Nonce<br>(ii) Gasprice<br>(iii) Data<br>(iv) Init | 7 | L3 | CO2/PO1 |
| 6(b) | Illustrate the concept of GAS in Ethereum | 7 | L3 | CO2/PO1 |
| 6(c) | Analyze the relationship between transaction, transaction trie and block header. | 6 | L4 | CO3/PO2 |
| | **OR** | | | |
| 7(a) | Illustrate the working of ethereum virtual machine with diagram. | 7 | L3 | CO2/PO1 |
| 7(b) | Illustrate different types of Accounts in ethereum Blockchain. | 7 | L3 | CO2/PO1 |
| 7(c) | Analyze the process of mining to validate and verify the blocks. | 6 | L4 | CO3/PO2 |

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# VIDYAVARDHAKA COLLEGE OF ENGINEERING

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## Second Semester M. Tech. Examinations: October 2024

## Blockchain Technology

**Duration: 3 Hour** | **Max. Mark: 100**

**INSTRUCTION TO STUDENTS**

**1) Answer One Full question from each module.**
**2) Three module questions are compulsory and the remaining two modules will have internal choice.**

| Q.No | Module-I | Marks | BL | CO's |
|---|---|---|---|---|
| 1 (a) | Explain the concepts of distributed systems and CAP theorem in blockchain. | 10 | L2 | CO1 |
| 1 (b) | Explain the various functions of blockchain. | 10 | L2 | CO1 |
| | **OR** | | | |
| 2 (a) | Explain various consensus algorithms in detail. | 10 | L2 | CO1 |
| 2(b) | Explain various types of blockchains. | 10 | L2 | CO1 |
| | **Module-II** | | | |
| 3 (a) | Classify the types of symmetric ciphers. | 4 | L2 | CO1 |
| 3(b) | Illustrate how to achieve fully decentralized ecosystem in blockchain. | 6 | L3 | CO2 |
| 3(c) | Apply AES algorithm to encrypt and decrypt for locally stored data using openssl and explain four stages. | 10 | L3 | CO2 |
| | **Module-III** | | | |
| 4 (a) | Apply RSA cryptosystem, a particular A uses two prime numbers, 13 and 17, to generate the public and private keys. If the public of A is 35. Find the private key of A. | 7 | L3 | CO2 |
| 4(b) | Illustrate the operation of pre-processing and Hash-computation in SHA-256. | 8 | L3 | CO2 |
| 4 (c) | Compare three security properties of hash functions. | 5 | L4 | CO3 |
| | **Module-IV** | | | |
| 5 (a) | Explain base58 encoding and vanity address. | 4 | L2 | CO1 |
| 5(b) | Illustrate various opcodes used in bitcoin development | 8 | L3 | CO2 |
| 5 (c) | Apply the structure of block and block header to explain how the transaction are stored in blockchain. | 8 | L3 | CO2 |
| | **Module-V** | | | |
| 6 (a) | Compare longest and heaviest chain in consensus mechanism. | 5 | L3 | CO2 |
| 6(b) | Analyze contract create transaction and message call transaction parameters and EVM operation. | 15 | L4 | CO3 |
| | **OR** | | | |
| 7 (a) | Compare Gas and Fee schedule in Ether Mining. | 5 | L3 | CO2 |
| 7(b) | Inspect the key parameters are provided by the execution agent in EVM operation. | 8 | L4 | CO3 |
| 7 (c) | Inspect the relationship between transaction, transaction trie and block header. | 7 | L4 | CO3 |

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Stream & Block ciphers

Identity, wealth.
Storage
Comm'n
Computation

Blockchain
Hyper ledger.
Storage
IPFS chain DB.
Comm'n
meshnets, internet.

$$\begin{array}{r} 16 \\ \times 12 \\ \hline 32 \\ 16 \\ \hline 48 \end{array}$$

$$\begin{array}{r} 13 \\ \times 17 \\ \hline 91 \\ 13 \\ \hline 221 \end{array}$$

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21IS/CS724

# VIDYAVARDHAKA COLLEGE OF ENGINEERING

Autonomous Institute, Affiliated to Visvesvaraya Technological University, Belagavi
Gokulam 3rd Stage, Mysuru 570 002

## Seventh Semester B.E. Make-up Examinations: February 2025

## Blockchain Technology

**Duration: 3 Hour** **Max. Mark: 100**

**INSTRUCTION TO STUDENTS**

1) Answer **One Full** question from each module.
2) Three module questions are compulsory and remaining two modules will have internal choice.

| Q.No | Module-I | Marks | BL | CO/PO |
|---|---|---|---|---|
| 1(a) | Explain the concepts of distributed systems and CAP theorem in blockchain. | 10 | L2 | CO1/PO1 |
| 1(b) | Explain how blockchain accumulates the block. | 5 | L2 | CO1/PO1 |
| 1(c) | Explain the generic structure of Blockchain. | 5 | L2 | CO1/PO1 |
| | **OR** | | | |
| 2(a) | Explain any five types of consensus algorithms in detail. | 10 | L2 | CO1/PO1 |
| 2(b) | Explain any five types of Blockchains. | 5 | L2 | CO1/PO1 |
| 2(c) | Explain any five applications of Blockchain | 5 | L2 | CO1/PO1 |
| | **Module-II** | | | |
| 3(a) | Explain the generic encryption and decryption model with a neat diagram. | 4 | L2 | CO1/PO1 |
| 3(b) | Illustrate how to achieve fully decentralized ecosystem in blockchain. | 6 | L3 | CO2/PO1 |
| 3(c) | Illustrate AES algorithm with a neat diagram | 10 | L3 | CO2/PO1 |
| | **Module-III** | | | |
| 4(a) | Illustrate Merkle tree with an example. | 5 | L3 | CO2/PO1 |
| 4(b) | Illustrate Asymmetric model used for message authentication and validation purposes. | 5 | L3 | CO2/PO1 |
| 4(c) | Analyze the working principle of RSA Algorithm for the given prime numbers 3 and 11 generate the public key and private key and encrypt the message "Hello". | 10 | L4 | CO3/PO2 |
| | **Module-IV** | | | |
| 5(a) | Define wallet and explain any 4 types of wallets that are used to store private keys. | 6 | L2 | CO1/PO1 |
| 5(b) | Analyze the transaction life cycle in bitcoin to transfer 20 bitcoins from sender to recipient. | 9 | L4 | CO3/PO2 |
| 5(c) | Analyze the working principles of various opcodes used in bitcoin reference client source code with an example. | 5 | L4 | CO3/PO2 |
| | **Module-V** | | | |
| 6(a) | Illustrate the various fields of transaction with an example. | 8 | L3 | CO2/PO1 |
| 6(b) | Illustrate Ethereum Stack with a neat diagram. | 6 | L3 | CO2/PO1 |
| 6(c) | Analyze the relationship between account trie, accounts, world state and state root hash with a neat diagram. | 6 | L4 | CO3/PO2 |
| | **OR** | | | |
| 7(a) | Illustrate the Ethereum virtual machine and its operation. | 8 | L3 | CO2/PO1 |
| 7(b) | Illustrate the relationship between transaction, transaction trie and Blockheader. | 6 | L3 | CO2/PO1 |
| 7(c) | Analyze the Scenario, "Sender address is 4718bf7a and recipient address is 74F7A2 and sender sends 2 ethers to recipient". Using state transition function. | 6 | L4 | CO3/PO2 |

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# VIDYAVARDHAKA COLLEGE OF ENGINEERING

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Gokulam 3rd Stage, Mysuru 570 002

QP 2/11/23 8/9

## Second Semester M. Tech. Examinations: October/November-2023

## Blockchain Technology

Duration: 3 Hour | Max. Mark: 100

INSTRUCTION TO STUDENTS

1) Answer **One Full** question from each module.
2) Three module questions are compulsory and the remaining two modules will have internal choice.

| Q.No | | Marks | BL | CO's |
|---|---|---|---|---|
| | **Module-I** | | | |
| 1 (a) | What is blockchain & explain the generic elements of blockchain | 10 | L2 | CO1 |
| 1 (b) | Describe the features and limitations of blockchain | 10 | L2 | CO1 |
| | **OR** | | | |
| 2 (a) | Explain the different types of blockchain and consensus requirement of blockchain | 10 | L2 | CO1 |
| 2(b) | Discuss the following in detail:<br>i) CAP theorem<br>ii) Byzantine General Problem | 10 | L2 | CO1 |
| | **Module-II** | | | |
| 3 (a) | What is decentralization? and classify the different platforms for decentralization in blockchain | 4 | L3 | CO1 |
| 3(b) | Explain the operations of stream and block ciphers in symmetric cryptography blockchain | 6 | L2 | CO2 |
| 3(c) | Apply the concepts of full ecosystem decentralization, in order to achieve complete decentralization. | 10 | L3 | CO2 |
| | **Module-III** | | | |
| 4 (a) | Apply RSA algorithm, to encrypt plain text into cipher text using blockchain | 8 | L3 | CO2 |
| 4(b) | Illustrate SHA-256 and SHA- 3 to create fixed length digest used in blockchain | 7 | L3 | CO2 |
| 4(c) | Compare Markel trees and Patricia trees used to store a dataset in blockchain | 5 | L4 | CO3 |
| | **Module-IV** | | | |
| 5 (a) | Explain keys and address in bitcoin. | 4 | L2 | CO1 |
| 5(b) | Define transaction and illustrate the transaction life cycle in bitcoin | 8 | L3 | CO2 |
| 5(c) | Illustrate the various tasks of bitcoin miners and explain steps of mining algorithm in detail | 8 | L3 | CO2 |
| | **Module-V** | | | |
| 6 (a) | Illustrate various components in Ethereum stack | 5 | L3 | CO2 |
| 6(b) | Analyze the structure of Ethereum virtual machine | 10 | L4 | CO3 |
| 6(c) | Analyze smart contracts and four precompiled contracts in Ethereum | 5 | L4 | CO3 |
| | **OR** | | | |
| 7 (a) | Analyze the following in detail:<br>i) Genesis block<br>ii) Currency (ETH and ETC)<br>iii) Forks and Gas | 15 | L4 | CO3 |
| 7(b) | Illustrate world state and account state in Ethereum blockchain | 5 | L3 | CO2 |

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